

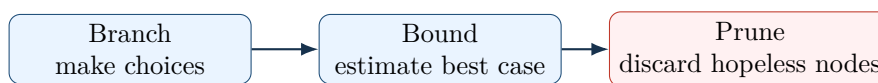
Branch and Bound Fundamentals

1 Core Idea

Branch and Bound

Branch and Bound is an exact optimization technique that searches a state-space tree while using bounds to avoid exploring branches that cannot improve the best solution already found.

It is used for optimization problems where the goal is to find the best feasible solution, not merely any feasible solution.

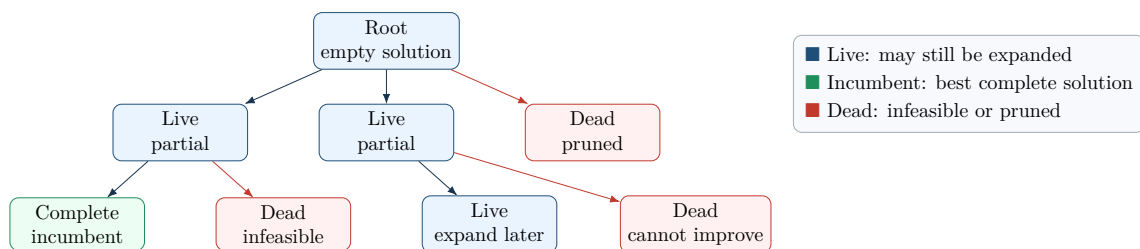


Not Greedy

Choosing the currently best-looking live node only controls the search order. Branch and Bound remains exact because other live nodes are kept until their bounds prove they can be discarded.

2 State-Space Tree

A state-space tree represents all possible decision paths. Branch and Bound keeps some generated nodes live and kills others using feasibility and bound tests.



Vocabulary

Live node	generated node that may still be expanded,
Dead node	completed, infeasible, expanded, or pruned node,
Incumbent	best complete feasible solution found so far.

3 Bounds

A bound is an optimistic estimate. It does not need to be the actual final value below the node; it must be good enough to prove when a branch cannot improve the incumbent.

3.1 Minimization Problems

For minimization, use a lower bound:

$$LB(N) = \text{least cost that node } N \text{ could possibly achieve.}$$

Let bestCost be the cost of the best complete feasible solution already found.

Minimization Pruning Rule

$$LB(N) \geq \text{bestCost} \implies \text{prune node } N.$$

Reason: even the most optimistic cost below N is not better than the current best cost.

3.2 Maximization Problems

For maximization, use an upper bound:

$$UB(N) = \text{greatest profit that node } N \text{ could possibly achieve.}$$

Let maxProfit be the profit of the best complete feasible solution already found.

Maximization Pruning Rule

$$UB(N) \leq \text{maxProfit} \implies \text{prune node } N.$$

Reason: even the most optimistic profit below N cannot beat the current best profit.

4 Lower Bound vs Upper Bound

Objective	Bound Used	Prune Condition
Minimize cost	Lower bound LB	$LB \geq \text{bestCost}$.
Maximize profit	Upper bound UB	$UB \leq \text{maxProfit}$.

Common Confusion

A lower bound is used for minimization, not maximization. An upper bound is used for maximization, not minimization. The bound must be optimistic in the direction of the objective.

5 Search Order

The live-node set may be managed in different ways.

Strategy	Live Node Removed Next
FIFO or BFS	The oldest generated live node.
LIFO or DFS	The most recently generated live node.
Least-cost search	The live node with the smallest lower bound.
Best-bound search	The live node with the most promising bound.

Breadth-first exploration is often introduced first, but many practical Branch and Bound algorithms choose the next live node by bound value.

6 Generic Algorithm

Branch and Bound Template

```
BranchAndBound(problem):
    create root node
    compute root bound
    liveNodes = {root}

    bestSolution = none
    bestValue = infinity      for minimization
                      = -infinity for maximization

    while liveNodes is not empty:
        node = remove one live node

        if node cannot improve bestValue:
            discard node
            continue

        if node is a complete feasible solution:
            update bestSolution and bestValue
            continue

        create child nodes

        for each child:
            if child is infeasible:
                discard child
            else:
                compute child bound
                if child can still improve bestValue:
                    add child to liveNodes

    return bestSolution
```

7 Small Numerical Illustration

Minimization Example

Suppose the current best complete solution has cost

$$\text{bestCost} = 18.$$

Three live nodes have lower bounds

$$LB(A) = 12, \quad LB(B) = 18, \quad LB(C) = 23.$$

Decision

For a minimization problem, prune when

$$LB \geq \text{bestCost}.$$

Node A :

$$LB(A) = 12 < 18,$$

so A may still lead to a better solution.

Node B :

$$LB(B) = 18 \geq 18,$$

so B cannot improve the incumbent.

Node C :

$$LB(C) = 23 \geq 18,$$

so C also cannot improve the incumbent.

Result

Keep A , prune B and C .

8 Correctness Intuition

Branch and Bound is correct when the bound is valid. A valid minimization lower bound never overstates how cheap a completion can be. A valid maximization upper bound never understates how profitable a completion can be.

1. Every feasible complete solution belongs to some branch of the state-space tree.
2. A branch is pruned only when its best possible optimistic value cannot improve the incumbent.
3. Therefore, no branch containing a better solution is discarded.

9 Complexity

Branch and Bound can greatly reduce practical search, but its worst-case complexity is still exponential or factorial for many optimization problems. The exact bound depends on the problem.

Problem Type	Typical Worst Case
Subset-style decisions	$\mathcal{O}(2^n)$.
Permutation-style decisions	$\mathcal{O}(n!)$.

Main Takeaway

Good bounds reduce the number of nodes expanded, but they do not automatically make an NP-hard problem polynomial in the worst case.

10 Final Review

Branch and Bound Essentials

Branch: split into decision subproblems.
Bound: estimate the best possible value below a node.
Incumbent: best complete feasible solution found so far.
Minimization: prune when $LB \geq \text{bestCost}$.
Maximization: prune when $UB \leq \text{maxProfit}$.